Modeling repeated evolution in HSC population

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Theorhetical modeling framework

Load libraries

```
library(tidyverse)
library(pbapply)
library(patchwork)
library(cowplot)
library(viridis)
```

Set parameters

Parameter values obtained from empirical estimates in Mitchell et al. 2022. Human genome size is a rough approximation.

```
hsc_pop <- c(80000,100000,120000)
ages <- seq(1,80,1)
annual_rates <- 2^seq(4,13)
diploid_genome_size <- 6.4e9
```

Run models

Code block will compute $P(X > 1)$ using Poisson and Binomial distributions (both results are identical). Note that these model the probability of a mutation occurring at the same site in any > 1 genomes in the population, disregarding the resulting state of the mutation.

```
## Poisson model
p.df <- pblapply(hsc_pop, function(x) {
  lapply(annual_rates, function(y) {
    df <- ppois(q = 1,
                lambda = ((y / diploid_genome_size) * x * ages),
                lower.tail = F) %>%
      as.data.frame() %>%
      rownames_to_column(var="age") %>%
      mutate(age = as.numeric(age)) %>%
      rename("p" = ".") # set the prob column name to p and convert rowname to age column

    df$N <- x # set HSC population size for df
    df$annual_muts <- as.factor(y) # set annual mutation rate for df
    df <- df %>% select(age, N, annual_muts, p)
    return(df)
  })
}) %>%
  bind_rows() %>%
```

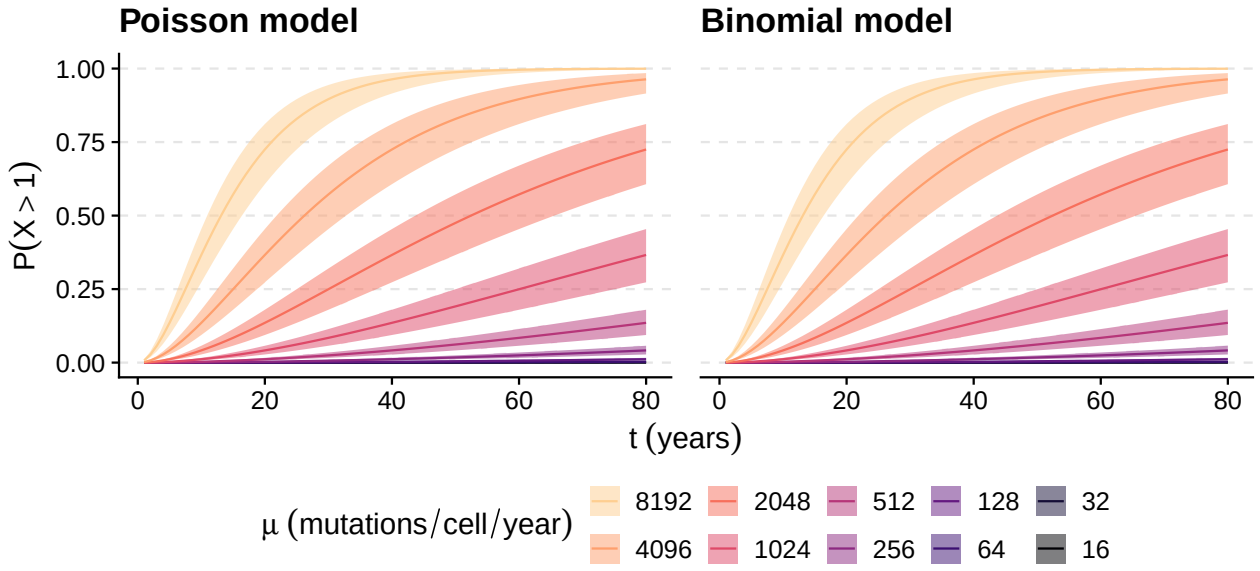
```r
  tibble()

## Binomial model
b.df <- pblapply(hsc_pop, function(x) {
  lapply(annual_rates, function(y) {
    df <- pbinom(q = 1,
                 prob = ((y / diploid_genome_size) * ages),
                 size = x,
                 lower.tail = F) %>%
      as.data.frame() %>%
      rownames_to_column(var = "age") %>%
      mutate(age = as.numeric(age)) %>%
      rename("p" = ".")

    df$N <- x
    df$annual_muts <- as.factor(y)
    df<-df %>% select(age, N, annual_muts, p)
    return(df)
  })
}) %>%
  bind_rows() %>%
  tibble()
```

Plot results

The plots below show curves as $P(X > 1)$ for the mean estimate for HSC population size, and ribbons show the range $\pm 20,000$ cells. Note that these are the conditional CDFs for the population and does not reflect sampling noise.



Implementing empirical estimates of μ

Estimate sources

Below, we will adapt the model above to compare the repeated evolution probabilities across classes of μ. The following values will be used:

- ***TERTp***: 3.8×10^{-7} (95% CI = 2.57×10^{-7} to 5.56×10^{-7}). However, this is computed from aggregating mutations across three distinct loci (-148, -124, and -57); thus we will divide these values by 3 to approximate a per-site estimate.
- **DNMT3A 882**: $1 \times 10^{-8} + 7.3 \times 10^{-9} + 9.9 \times 10^{-10}$ (R882H, R882C, and R882S, respectively). Estimated in MacGregor et al. 2025 (95% CI not provided).
- **Genome-wide average**: 2.6×10^{-9} (95% CI = 2.67×10^{-9} to 2.58×10^{-9}); estimated in Mitchell et al. 2022.

```
emp_mu <- data.frame(class = factor(c("TERTp", "DNMT3A 882", "Genome-wide"), levels = c("TERTp", "DNMT3
                     mean = c(3.8e-07 / 3, (1e-8 + 7.3e-9 + 9.9e-10), 2.6e-9),
                     ci_95high = c(5.566e-7 / 3, NA, 2.58e-9),
                     ci_95low = c(2.578e-7 / 3, NA, 2.67e-9))

emp_mu
```

```
##         class         mean    ci_95high     ci_95low
## 1       TERTp 1.266667e-07 1.855333e-07 8.593333e-08
## 2  DNMT3A 882 1.829000e-08           NA           NA
## 3 Genome-wide 2.600000e-09 2.580000e-09 2.670000e-09
```

Data & modeling

Because these site-wise values already accounts for the diploid human genome size ($2L \approx 6.4 \times 10^9$ loci), we can apply the Poisson CDF to estimate the probability of repeated evolution for each class of μ. Below, we will hold the population size constant at $N = 100000$ HSCs.

```
# Create long data frame for easy across-compute
emp.df <- emp_mu[rep(seq_len(nrow(emp_mu)), each = length(seq(0,80,0.1))),] %>%
  tibble()
emp.df$age <- rep(seq(0,80,0.1),
                  times = nrow(emp_mu))
emp.df <- emp.df %>%
  pivot_longer(cols = c("mean", "ci_95high", "ci_95low"),
               names_to = "quant",
               values_to = "mu") %>%
  filter(!is.na(mu))

# Compute probabilities using Poisson CDF
emp.df <- emp.df %>%
  mutate(p = ppois(q = 1,
                   lambda = (mu * 100000 * age),
                   lower.tail = F))

head(emp.df)
```

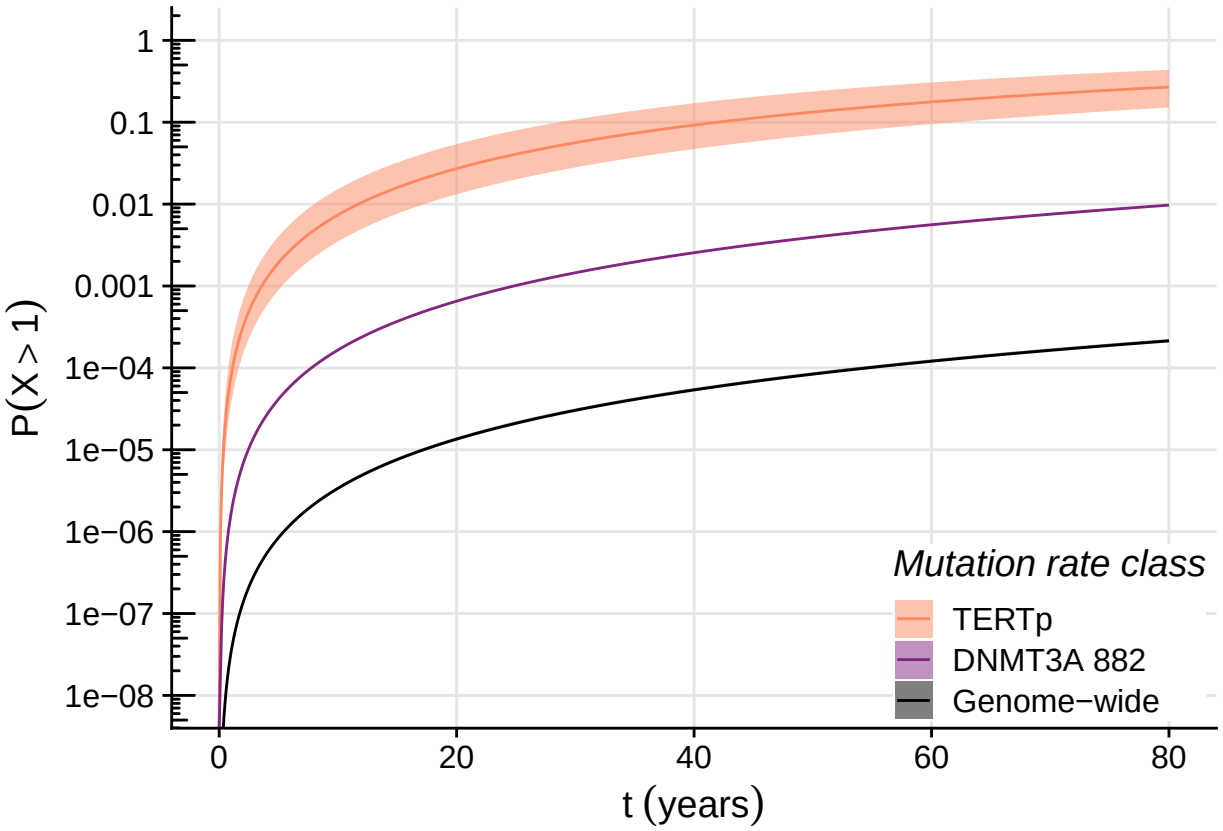
```
## # A tibble: 6 x 5
##   class   age quant               mu           p
##   <fct> <dbl> <chr>            <dbl>       <dbl>
## 1 TERTp   0   mean      0.000000127  0
## 2 TERTp   0   ci_95high 0.000000186  0
## 3 TERTp   0   ci_95low  0.0000000859 0
## 4 TERTp   0.1 mean      0.000000127  0.000000802
## 5 TERTp   0.1 ci_95high 0.000000186  0.00000172
## 6 TERTp   0.1 ci_95low  0.0000000859 0.000000369
```

Visualization and statistics

Here, we'll plot the results in log10 space to better visualize the differences across classes of μ.

```
# Remove mu column and pivot wider to accommodate visualizing confidence intervals
emp.df_plotting <- emp.df %>%
  select(!mu) %>%
  pivot_wider(names_from = quant, values_from = p)

# Plot curves
ggplot(data = emp.df_plotting, mapping = aes(x = age, color = class, fill = class)) +
  geom_hline(yintercept = 10^seq(-8,0,1),
             linetype = 1,
             linewidth = 0.5,
             color = "gray90") + # horizontal lines at log10 breaks
  geom_vline(xintercept = c(0,20,40,60,80),
             linetype = 1,
             linewidth = 0.5,
             color = "gray90") + # vertical lines at log10 breaks
  geom_ribbon(aes(ymin = ci_95low,
                  ymax = ci_95high),
              alpha = 0.5,
              color = "transparent") + # confidence intervals
  geom_line(aes(y = mean)) + # mean trend
  scale_y_continuous(trans = 'log10',
                     breaks = 10^seq(-8,1,1),
                     labels = 10^seq(-8,1,1)) +
  coord_cartesian(ylim = c(1e-8, 1)) +
  scale_color_viridis(option = "magma",
                      discrete = TRUE,
                      end = 0.75,
                      direction = -1) +
  scale_fill_viridis(option = "magma",
                     discrete = TRUE,
                     end = 0.75,
                     direction = -1) +
  theme_cowplot() +
  theme(legend.position = c(0.99,0.01),
        legend.justification = c(1,0),
        legend.background = element_rect(fill="white"),
        legend.title = element_text(face="italic"),
        axis.title = element_text(face="italic")) +
  labs(x = expression(t~(years)),
       y = expression(P(X>1)),
       color = "Mutation rate class",
       fill = "Mutation rate class") +
  annotation_logticks(side = "l")
```



Interpretation

From these data, we can also extract some key statistics. First, the estimated probability of repeated evolution at a single *TERTp* hotspot locus is $\approx 2.69e-01$. This value contrasts estimates for *DNMT3A R882* and genome-wide classes are $\mathrm{P} \approx 9.72e-03$ and $\mathrm{P} \approx 2.13e-04$, respectively.

To place into a comparative framework, the odds ratio of repeated mutation at a *TERTp* hotspot locus (relative to the genome-wide average mutation rate) is $1.73e+03$. While not modeled here, this elevated mutation rate likely suggests that repeated mutation at TERTp is present across human populations, but remains unobserved when the fitness advantage of the mutation is low (and/or when sampling is sparse).